

1. Administrative & Study Identification

Full Study Title: A Phase III, Multicenter, Double-Blind, Placebo-Controlled, Treat-Through Study to Assess the Efficacy and Safety of Induction and Maintenance Therapy with R07790121 in Patients with Moderately to Severely Active Ulcerative Colitis
Short Title/Acronym: AMETRINE-1 **Protocol Number:** GA45329 **NCT Number:** NCT06589986 **Sponsor Name:** Hoffmann-La Roche
Investigational Product: Afimkibart (also known as R07790121, an anti-TL1A monoclonal antibody) **Phase of Development:** Phase III
Medical Monitor: Hoffmann-La Roche Clinical Operations Lead (e.g., Anthony Lei, UK Contact)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of Afimkibart (R07790121) compared with placebo for induction and maintenance therapy in participants with moderately to severely active ulcerative colitis (UC). **Secondary Objectives:** To evaluate clinical remission by modified Mayo Score (mMS), endoscopic improvement/remission, histologic improvement, changes in partial Mayo score over time, and the overall safety and tolerability profile of Afimkibart.

2.2 Background & Rationale To address the significant unmet medical need in patients with moderately to severely active ulcerative colitis who have failed prior conventional or advanced therapies. TL1A is a protein naturally found in the body that regulates gut mucosal immunity and inflammation. TL1A levels and activity are elevated in UC patients, driving disease development and worsening. Afimkibart is designed to inhibit TL1A, providing a novel mechanism to achieve sustained disease control, clinical remission, and an improved long-term prognosis.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Treat-Through Study) **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 60% chance of assignment to the Afimkibart group, 40% chance of assignment to the placebo group.

3.2 Planned Enrollment & Duration Target \$N\$: 400 participants
Number of Sites: Multicenter (Global investigational sites)
Estimated Study Start: 17-Sep-2024 **Estimated Primary Completion:** 30-Jan-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 16 to 80 years. 2. Confirmed diagnosis of moderately to severely active Ulcerative Colitis with a body weight ≥ 40 kg. 3. Demonstrated inadequate response, loss of response, and/or intolerance to at least one conventional therapy (aminosalicylates, corticosteroids, and/or immunosuppressants) OR prior advanced therapy (biologics or targeted small molecules like anti-TNF, anti-IL12/23, anti-integrin, S1P receptor modulators, or JAK inhibitors).

4.2 Exclusion Criteria 1. Pregnant or planning to become pregnant during the study through 95 days after the final dose of the drug. 2. Current diagnosis of other forms of colitis (e.g., indeterminate colitis, microscopic colitis, ischemic colitis, infectious colitis, radiation colitis, Crohn's disease) or active diverticular disease. 3. Presence of UC complications such as fulminant colitis, toxic megacolon, symptomatic bowel strictures, primary sclerosing cholangitis, or patients with an ostomy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - **Induction Phase:** Administered on Day 1, Week 2, and every 4 weeks until Week 10. - **Maintenance Phase:** Administered every 4 weeks from Week 12 to Week 52. **Route of Administration:** Intravenous (IV) infusion during the Induction Phase; Subcutaneous (SC) injection during the Maintenance Phase. **Duration of Treatment:** 52 weeks for the main treat-through period, with participation lasting up to approximately 70 weeks including an optional Open-Label Extension Phase and follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of certain baseline conventional UC treatments as specified by the protocol. **Prohibited:** Concurrent use of other advanced targeted therapies (biologics, JAK inhibitors, S1P modulators) or live vaccines during the trial.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Up to Day 0	Informed Consent, Medical History, Baseline Endoscopy/Biopsy, Randomization
2	Weeks 0, 2, 6, 10	IV Infusion of Study Drug/Placebo, Safety Labs, Symptom Monitoring
3	Weeks 12 to 52	SC Injection every 4 weeks, Efficacy Assessment, QoL Questionnaires
4	Week 56 / Follow-up	Final Endoscopy, Clinical Remission Assessment, Exit Interview, Option for Open-Label Extension

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Clinical remission at a specified time point (e.g., Week 14 or 56), defined by the modified Mayo Score (mMS). **Measurement Method:** Clinical assessment (stool frequency, rectal bleeding) combined with Central Endoscopy Reading (endoscopic subscore).

7.2 Secondary & Exploratory Endpoints 1. Endoscopic improvement and endoscopic remission. 2. Histologic improvement and combined histologic-endoscopic mucosal improvement. 3. Safety outcomes including the incidence and severity of adverse events.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via site visits every 4 weeks and continuous patient reporting. **Serious Adverse Events (SAEs):** Reported to the sponsor and regulatory authorities within 24 hours of site awareness. **Data Monitoring Committee (DMC):** External independent Data Monitoring Committee to evaluate ongoing safety and benefit-risk profiles.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcomes; Safety Set for all participants receiving at least one dose. **Sample Size Justification:** Target $N=400$ provides adequate statistical power ($\alpha = 0.05$) to detect a clinically meaningful difference in clinical remission rates between the Afimkibart and placebo arms. **Handling of Missing Data:** Non-Responder Imputation (NRI) for categorical efficacy endpoints (e.g., clinical remission).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring to ensure protocol adherence and data integrity. **Audit Trail:** Secure electronic Case Report Form (eCRF) data entry with full audit trails and central reading of endoscopic/histologic samples to minimize bias.

11. Study Identifiers & Governance

Primary ID: GA45329 **Registry Number:** NCT06589986 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** AMETRINE-1 **Official Regulatory Title:** A Phase III, Multicenter, Double-Blind,

Placebo-Controlled, Treat-Through Study to Assess the Efficacy and Safety of Induction and Maintenance Therapy With R07790121 in Patients With Moderately to Severely Active Ulcerative Colitis

12. Clinical Framework (Ulcerative Colitis Specific)

Primary Condition: Moderately to Severely Active Ulcerative Colitis **Diagnosis Threshold:** Endoscopically confirmed disease activity and inadequate response to prior therapies **Medical Methodology:** Anti-TL1A monoclonal antibody therapy (Afimkibart) **Optimization Target:** Clinical and endoscopic remission (Mucosal healing)

13. Study Procedures & Methodology

Management Pathway: Treat-through design evaluating continuous induction and maintenance efficacy **Education Protocol (HCP):** Site initiation visits, training on SC injection administration, and central endoscopy reading standards **Education Protocol (Patient):** Informed consent process, disease education, and SC self-administration training (if applicable) **Quality Audit Mechanism:** Centralized evaluation of endoscopic videos and biopsy samples

14. Observation & Follow-Up Schedule

Total Duration: Approximately 70 weeks (including follow-up) **Onsite Visit Cadence:** Day 1, Week 2, then every 4 weeks during treatment **Remote Monitoring Frequency:** Standard safety check-ins between dosing visits **Checklist Review Interval:** Every 4 weeks (Q4W) preceding each dose administration

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				